Global Mart Sales Performance Analysis Report

Problem Statement

Global Mart is a multinational retailer selling Furniture, Office Supplies, and Technology across several U.S. regions. The company is experiencing fluctuations in sales performance such as a 6% decline in year-over-year revenue between 2023 and 2024, uneven regional performance, product level inconsistencies, revenue losses and a drop in orders by 41 likely caused by late deliveries with a shipping performance of 39.1%. Leadership needs deeper insights into sales trends, customer behavior, product profitability, regional performance and operational efficiency to make informed, data-driven decisions.

The goal is to provide Global Mart with actionable insights and recommendations highlight growth opportunities, identify problem areas, and support strategic decision making through exploring, analyzing, and visualizing the company's sales data.

Project Objectives

The primary objectives of this report are to transform Global Mart's raw sales data into coherent, insight driven analytical insights and recommendations that supports data-driven decision-making across all business units. Specifically, the report aims to;

1. Provide insights into overall business performance
    I. Evaluate total revenue, profit, orders, and profit margin
    II. Identify annual growth trends, including YoY performance for 2022–2024
    III. Present high-level KPIs that reflect the company's financial health
2. Analyze product-level performance
    I. Identify top performing and bottom performing products and sub-categories
    II. Examine sales, profit, and order patterns across product categories
    III. Detect outliers that distort performance metrics
3. Understand customer and order behavior
    I. Analyze order value patterns, discount usage, and quantity per order

II. Define order-value bands and assess how they impact profitability

III. Provide insights into customer demand and purchasing trends

4. Evaluate regional sales performance

I. Compare sales, profit, and order trends across regions and states

II. Identify high-performing and underperforming states

III. Assess shipping efficiency, including days to ship and late delivery impacts

5. Apply time intelligence to reveal trends over time

I. Analyze month-to-date (MTD), year-to-date (YTD), and quarter-to-date performance

II. Compute year-over-year (YoY) growth, rolling averages, and seasonal trends

III. Highlight periods of strong or weak sales performance and growth drivers

6. Present actionable insights through interactive dashboards

I. Build dashboards for Executive Summary, Sales Overview, Product Analysis, Customer Analysis, Regional Performance, and Time Intelligence

II. Enable drill-downs, filters, slicers, and user-friendly visual interactions

III. Consolidate insights to support strategic planning, pricing decisions, inventory management, and regional operations

Methodology for the analysis

1.Data Import and Transformation

The analysis was conducted using the Global Mart Sales dataset provided in Excel format. The dataset contained sales transactions, product information, order details, geographic details, shipping data, and date fields. All data was imported into Power BI for modeling and report development.

Several steps were taken to ensure data quality, accuracy, and usability through cleaning and transformation such as;

1) Converted date fields into proper date formats using locale
2) Ensured numeric fields (Sales, Profit, Quantity, Discount) were correctly typed
3) Checked for missing, duplicate, or inconsistent entries. The null values and outliers were filtered out for the sake of data quality and accuracy

4) Errors were removed and text was standardized in the state column.
5) Created a custom region table with the states correctly assigned to regions to correct inconsistent region assignments.
6) Created fact and dimension tables for data modelling and created primary keys for the dimension tables then added them to the fact table.
7) Created a date table called calender_dim and marked it as the official Date table to enable accurate time intelligence functions.

2. Data Modeling

Opted to go with a star schema model to ensure optimal performance and analysis. A one-to-many relationship from each dimension to the fact table was established and ensured single direction filters for performance and clarity

3.KPI Calculation and DAX measures

Multiple DAX measures was created to support KPIs and analytical insights including;

I. Core sales metrics
II. Product metrics
III. Customer and order metrics
IV. Regional performance metrics
V. Time intelligence and trends
VI. Advanced business metrics
VII. 3 calculated tables for deeper insights and analysis

4. Visualization Development

Six interactive dashboards with slicers and filters were designed to deliver clear, actionable insights including;

I. Executive overview with KPIs and charts showing overall business health
II. Sales overview showing analysis of revenue, orders, profit, discounting behavior, sales trends, and category insights.
III. Product analysis showing profitability, revenue contribution, order frequency, outliers, and sub-category performance
IV. Customer analysis showing average quantity per order, discount patterns, order bands, and purchase behavior.
V. Regional performance showing state and region comparisons, shipping performance, late delivery impact, and market strengths.
VI. Time intelligence showing seasonal trends, and historical patterns across 2022–2024.

Insights

Sales and Profitability Insights

1. Declining Revenue Despite High Profitability between 2023 and 2024

- Total revenue of $469.35K with a strong 69.1% profit margin indicates good pricing power but volume challenges
- Revenue declined 6% YoY (2023-2024), suggesting market share loss or reduced customer demand
- Revenue YTD ($227.43K) is lower than last year's comparable period ($241.92K), confirming the downward trend

2. Category Performance Disparity

- Office Supplies leads revenue at $165K, followed by Technology ($155K) and Furniture ($149K)
- However, Office Supplies shows the steepest decline at -21.4% YoY growth, requiring immediate attention
- Technology also declining at -2.1% YoY, while Furniture shows positive growth at +9.7%

3. Discount Rate is a majorly affects revenue

- When average discount rate increases by 0.01, total revenue decreases by $23.19K
- This is identified as the key influencer of revenue decline
- Current discounting strategy is eroding revenue without driving sufficient volume

Product Performance Insights

4. Top Products Drive Disproportionate Value

- Top 10 products generate $17,157.33 in revenue (3.7% of total) with $11,897.55 in profit
- Leading products such as Laptops front ($1,967.58), Pens represent ($1,869.25), and Accessories fear ($1,854.92)

5. Bottom Products Drain Resources

- Bottom 10 products generate only $20.46 to $35.10 in revenue each
- Products like "Binders social" ($20.46), "Tables past" ($20.54), and "Bookcases firm" ($24.41) are underperforming

6. Sub-Category Leaders

- Pens ($69K), Laptops ($57K), and Chairs ($54K) are top revenue-generating sub-categories
- These three sub-categories also lead in total profit and order volume

Customer & order behavior insights

7. Order volume declining significantly

- Orders dropped from 488 last year to 447 this year (-41 orders, -8.4% decline)
- Average revenue per order remains healthy at $501.98, indicating the issue is volume, not value

8. RFM analysis reveals at-risk states

- Texas has the lowest RFM-Total score (3) with lowest recency, frequency, and monetary values
- Pennsylvania (RFM: 20) and Michigan (RFM: 18) are the most engaged customers
- Texas customers haven't ordered recently (11/14/2024) and have lowest order frequency (103) and revenue ($51,692)

9. Order frequency varies significantly by state

- Pennsylvania leads with 135 orders, while Texas lags at 103 orders
- California (111), Florida (111), and Ohio (111) show similar moderate performance

Regional performance insights

10. Regional revenue distribution is unbalanced

- West region dominates with 37.86% of total orders (354 orders)
- Central region significantly underperforms at only 11.87% (111 orders)
- South (27.38%, 256 orders) and East (22.89%, 214 orders) show moderate performance

11. State-level performance gap is substantial

- Top performer Michigan ($65,021) outperforms worst performer Texas ($51,692) by $13,329 (25.8% difference)
- Mid-tier performers cluster around $55-65k range

Operational & shipping insights

12. Shipping performance crisis

- Shipping performance status shows "Needs Improvement" at 39.1% failure rate

- This means nearly 4 out of 10 shipments are experiencing delays or issues

13. Late deliveries causing significant revenue loss

- Every state experience $30-38k in late delivery revenue loss
- Pennsylvania, Michigan, and California each lose $38k (highest losses)
- Total late delivery losses could exceed $250k+ across all states

14. Shipping days vary by region

- Profit distribution across 0-9 days to ship shows inconsistent patterns
- Some regions show profit spikes at 3-4 days, others at 6-7 days

Time-based insights

15. Seasonal revenue patterns

- Revenue peaks in months 6-8 (June-August) with $46K, $38K, and $39K
- Significant drop in early months (February: $12K, April: $19K)

16. Quarterly distribution shows q2 dominance

- Quarter 2 leads with 27.37% of annual revenue
- Quarter 1 is weakest at 22.4%
- Quarter 3 (25.55%) and Quarter 4(24.68%) show moderate performance

17. Year-end forecast shows modest recovery

- Revenue forecast for year-end is $246.33K
- This represents slight improvement from current YTD but still below targets

Recommendations

Based on the comprehensive analysis of Global Mart's sales performance across products, customers, regions, and operational metrics, the following strategic recommendations are proposed to improve revenue growth, profitability, and operational efficiency.

1. Strengthen high-performing product categories

- Focus inventory and marketing efforts on Pens, Laptops, and Chairs, which consistently generate the highest revenue and profit.
- Expand product bundles combining top-performing items (e.g., "Office Starter Kits" with pens, paper, and accessories).
- Negotiate better bulk purchasing deals to maintain strong profit margins.

2. Address low-performing products

- Review pricing strategies and promotional plans for low-revenue items such as Binders, Bookcases, and low-selling laptop models.
- Phase out persistently underperforming SKUs to reduce storage and warehousing costs.
- Reinvest savings into high-demand products.

3. Improve customer retention through engagement

- Target high-value customer regions (e.g., Michigan, Pennsylvania) with loyalty discounts or exclusive deals.
- Engage at-risk customers using personalized offers based on RFM segmentation.
- Encourage more frequent orders by promoting subscription options for office supplies.

4. Optimize regional sales strategy

- Prioritize sales and marketing investment in top-performing regions such as the East and Central regions.
- Investigate underlying issues in low-performing states (e.g., Texas) by reviewing customer satisfaction, product availability, and delivery times.
- Tailor product mix per region to match demand patterns and category preferences.

5. Reduce shipping delays to minimize revenue loss

- Shipping delays are contributing to significant revenue loss (up to $38K per state).
- Improve logistics processes by partnering with faster carriers or optimizing internal fulfillment.
- Implement delivery performance dashboards to monitor and address late shipments proactively.

6. Leverage time intelligence insights

- With declining year on year revenue (-6%), implement monthly performance reviews to intervene early.
- Use forecasting outputs to adjust sales targets and inventory planning.
- Align promotional activities with months historically showing lower sales.

7. Enhance discount strategy

- Since high discounts are negatively correlated with revenue tighten discount policies.
- Offer targeted promotions to specific customer groups rather than broad discounts.
- Analyze discount impact per product to avoid margin erosion.